Assignment 1.5

Day 5: Assemble and Disassemble a PC Practical (1 hour): Full hands-on session: Assemble and disassemble a desktop PC Task: Document the process with labeled steps. Pending

Task – 1

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

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After opening the PC cabinet, the following main components can be found inside.

No.	Component	Short Description
1	**Motherboard**	The main circuit board of the computer. It connects the CPU, RAM, storage, expansion cards, and other components.
2	**CPU / Processor**	The main processing unit that executes instructions and performs calculatios for the computer.
3	**CPU Cooler / Fan**	Removes heat generated by the processor and keeps the CPU at a safe operating temperature.
4	**RAM**	Temporary memory used by the computer to store data and programs that are currently being used.
5	**Power Supply Unit (PSU)**	Converts AC electricity from the wall outlet into the DC power required by computer components.
6	**HDD / SSD**	Provides permanent storage for the operating system, applications, and user files.
7	**Graphics Card (GPU)**	Processes graphics and sends video output to the monitor. Some computers use integrated graphics instead.
8	**Cabinet / Case**	The enclosure that holds and protects the internal computer components.
9	**Case Fans**	Circulate air through the cabinet to help keep internal components cool.
10	**CMOS Battery**	A small battery on the motherboard that maintains BIOS/UEFI settings and system time when the PC is powered off.
11	**SATA Data Cable**	Connects SATA storage devices such as HDDs and SATA SSDs to the motherboard.
12	**Power Cables**	Supply electrical power from the PSU to components such as the motherboard, drives, and graphics card.

Task – 2

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard..

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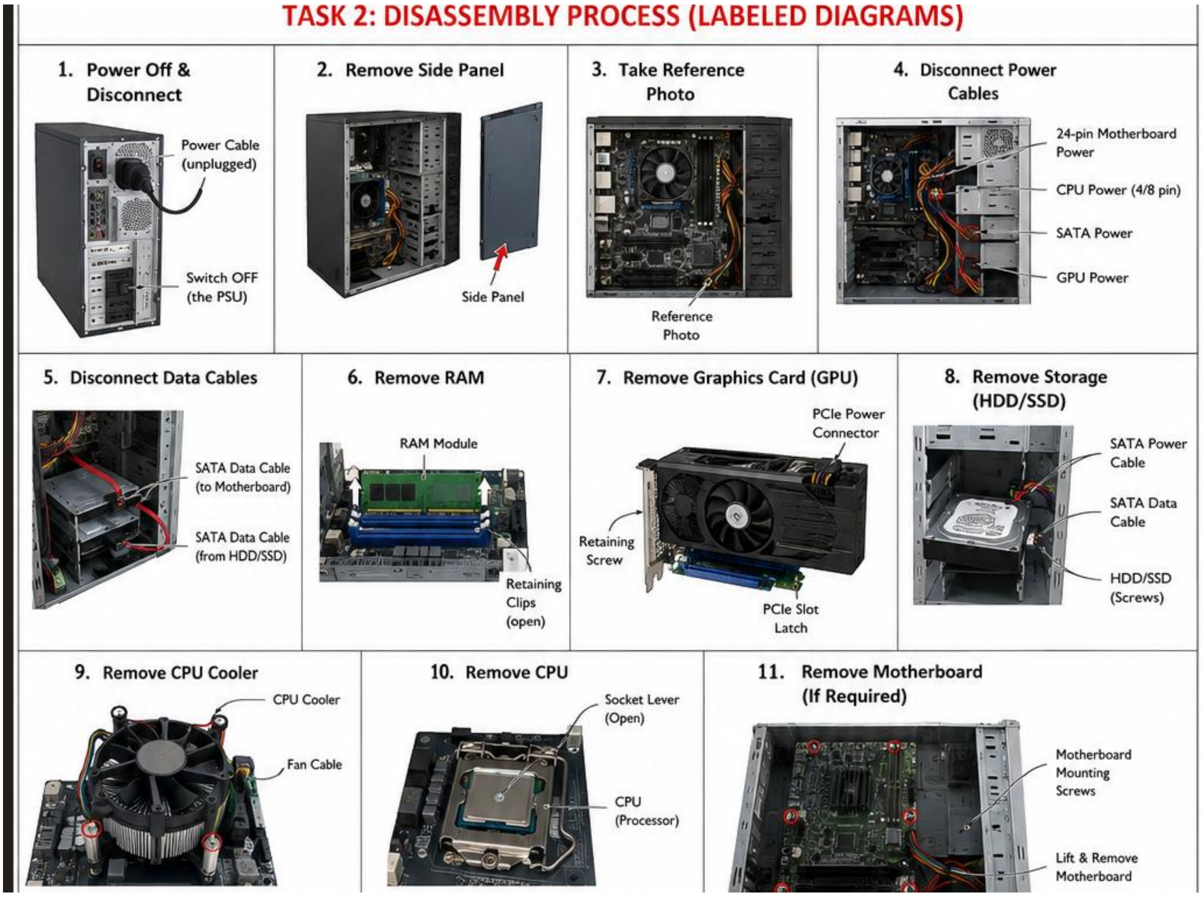
The following labeled diagrams illustrate the step-by-step process of safely disassembling a desktop computer



Step 1 – Power Off and Disconnect the PC: The computer was completely shut down and disconnected from the power source before starting the disassembly process.

Step 2 – Remove Side Panel: The side panel was removed to access the internal components of the desktop computer.

Step 3 – Internal Reference Photo: A photograph of the original cable and component arrangement was taken to ensure correct reassembly.

Step 4 – Disconnect Power Cables: Power cables were carefully disconnected from the components to prepare the PC for component removal.

Common cables include:

- 24-pin motherboard power
- 4/8-pin CPU power
- SATA power
- GPU power
- Fan power

Step 5 – Disconnect SATA Data Cable: The SATA data cables connecting the storage device to the motherboard were disconnected carefully.

Step 6 – Remove RAM: The RAM retaining clips were opened and the memory module was carefully removed by holding its edges.

Step 7 – Remove Graphics Card: The graphics card was disconnected from its power supply, released from the PCIe slot, and carefully removed.

Step 8 – Remove HDD/SSD: The storage device was disconnected and removed from its mounting position inside the cabinet.

Step 9 – Remove CPU Cooler: The CPU cooling assembly was carefully removed to expose the processor.

Step 10: Remove CPU

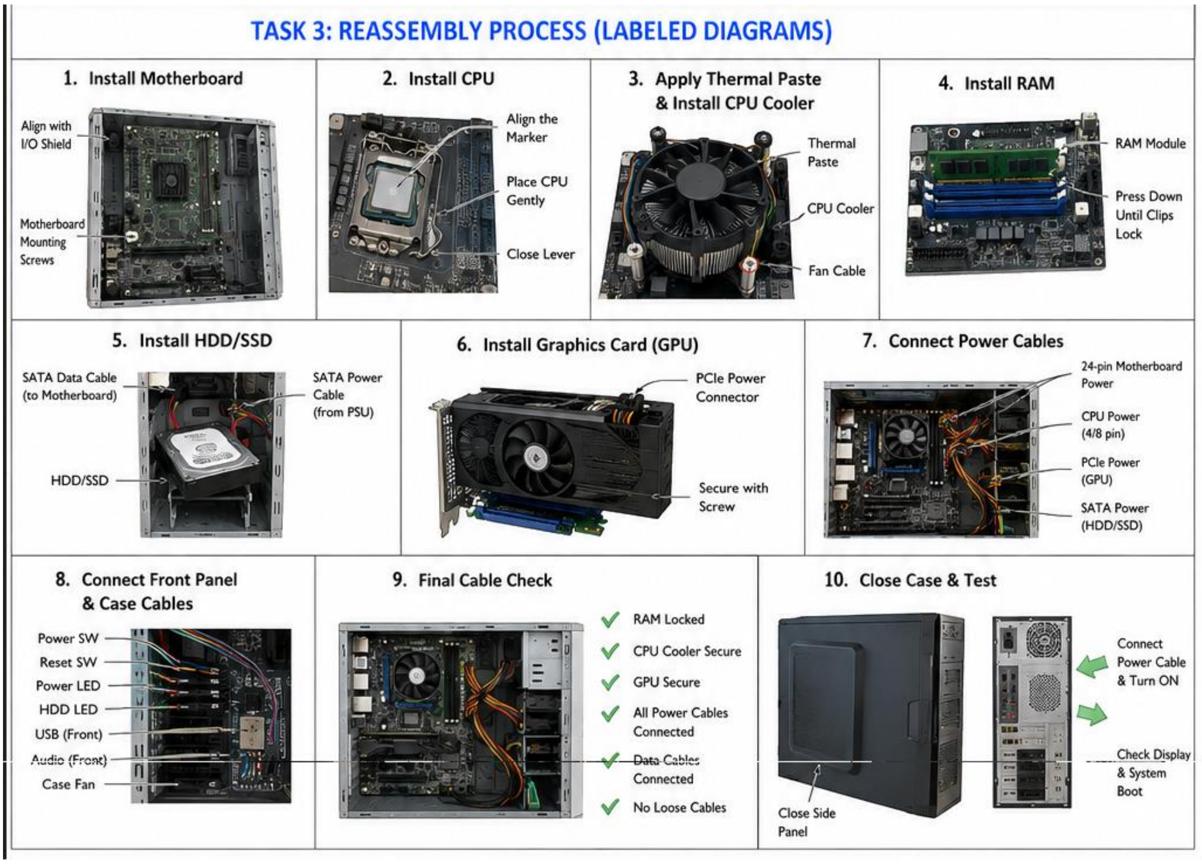
If instructed to remove the CPU:

1. Open the CPU socket retention mechanism.
2. Carefully lift the processor.
3. Hold it only by its edges.
4. Place it in a safe location.

Step 11 – Remove Motherboard: After disconnecting all cables and removing the mounting screws, the motherboard was carefully removed from the cabinet.

Task – 3

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.



Task – 4

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

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No.	Safety Precaution	Why It Is Important
1	**Switch off the PC**	Prevents electrical hazards and accidental component damage.
2	**Unplug the power cable**	Disconnects the PC from the electrical supply.
3	**Use an anti-static wrist strap**	Helps prevent electrostatic discharge from damaging sensitive components.
4	**Work on a clean, dry surface**	Prevents dust, moisture, and unwanted objects from damaging components.
5	**Hold components by their edges**	Prevents fingerprints and accidental contact with sensitive contacts.
6	**Do not force connectors**	Connectors should normally fit in the correct orientation; forcing them can cause damage.
7	**Keep screws organized**	Prevents losing screws and installing an incorrect screw in the wrong location.
8	**Do not touch electrical contacts**	Oils and dirt from fingers can contaminate contacts.
9	**Be careful with the CPU**	CPU pins/contacts can be easily damaged.
10	**Do not work on a powered PC**	Reduces risk of electrical shock and component damage.
11	**Keep liquids away**	Liquid can cause short circuits and permanent damage.
12	**Check cables before powering on**	Prevents incorrect connections and possible component damage.

Tool	Purpose
Phillips screwdriver **Anti-static wrist strap**	Used to remove and install most PC screws. Protects components from electrostatic discharge.
Small container/screw tray	Keeps screws organized during disassembly.
Flashlight	Helps inspect components and connectors inside the cabinet.
Compressed air / air blower	Can be used to remove dust, following appropriate precautions.

Tool	Purpose
Thermal paste	Used when reinstalling a CPU cooler if the existing paste needs replacement.
Cable ties	Used to organize cables during reassembly.
Clean workspace	Provides a safe area for handling components.